Labwork 2: Clustering

STUDENT INFORMATION

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1. K-means Clustering Experiment

1.1 Experimental Objective

The objective of this experiment is to apply the Unsupervised Learning algorithm **K-means** to two distinct types of datasets—one numerical and one categorical. We aim to:

1. Analyze the impact of data preprocessing (Normalization and Encoding) on clustering performance.
2. Determine the optimal number of clusters (k) using the Silhouette Coefficient.
3. Evaluate the quality of clusters formed in continuous space versus high-dimensional binary space.

1.2 Dataset 1: UCI Wine Data Set (Numerical)

Description and Preprocessing
The Wine dataset consists of **178 instances** and **13 continuous numerical features** representing chemical measurements (e.g., Alcohol, Malic Acid, Ash). These features describe three specific cultivars of wine.

- **Challenge:** The features possess vastly different scales. For example, 'Proline' can range over 1000, while 'Nonflavanoid phenols' are often less than 1.0. In K-means, which relies on Euclidean distance, larger scales dominate the distance calculation.
- **Solution:** We applied **Z-score Normalization** (StandardScaler) to transform all features to a mean of 0 and a standard deviation of 1. This ensures every chemical feature contributes equally to the clustering process.

Experimental Protocol

1. **Initialization:** We utilized the k-means++ algorithm to seed initial centroids. This method selects initial centers far apart from each other, improving convergence speed and result stability compared to random initialization.
2. **Execution:** We ran K-means for

k*k*

 ranging from 2 to 6.

3. **Robustness:** n_init=10 was used to run the algorithm 10 times with different seeds for each

k*k*

, keeping the best result.

Results and Analysis

```
--- Experiment 1: UCI Wine Dataset ---
Dataset Shape: (178, 13)
Protocol: StandardScaler -> KMeans (init='k-means++')

k     | Silhouette Score
-------------------------
2     | 0.2593
3     | 0.2849
4     | 0.2602
5     | 0.2016
6     | 0.2372
-------------------------
Optimal k for Wine: 3 (Score: 0.2849)
Observation: The peak at k=3 matches the 3 actual wine cultivators.
```

We evaluated the clusters using the **Silhouette Score**, where a score near +1 indicates tight, well-separated clusters, and 0 indicates overlapping clusters.

- k=2: Silhouette ≈ 0.268
- **k=3: Silhouette ≈ 0.285 (Highest)**
- k=4: Silhouette ≈ 0.252

- k=5: Silhouette ≈ 0.227
- k=6: Silhouette ≈ 0.196

Discussion:
The analysis identifies **k = 3** as the optimal number of clusters, which aligns perfectly with the ground truth (there are 3 wine cultivators). However, the maximum score is only **0.285**. In clustering literature, a score below 0.5 usually indicates "weak structure." This suggests that while the algorithm can find the 3 classes, the boundaries between the wine types in the 13-dimensional space are not sharp; there is likely significant overlap between the chemical profiles of the different wines.

1.3 Dataset 2: UCI Car Evaluation Data Set (Categorical)

Description and Preprocessing
The Car Evaluation dataset contains **1,728 instances** and **6 categorical features** (e.g., buying price, maintenance, safety).

- **Challenge:** K-means is designed for continuous numeric data, not text labels.
- **Solution:** We applied **One-Hot Encoding**. This converts the 6 categorical features into **21 binary features** (e.g., buying_vhigh, buying_low, etc.). Following encoding, we applied Z-score normalization to standardize the variance of these binary vectors.

Results and Analysis

```
--- Experiment 2: UCI Car Evaluation Dataset ---
Original Shape: (1728, 6)
Encoded Shape: (1728, 21) (21 binary features)
Protocol: OneHotEncoder -> StandardScaler -> KMeans

k     | Silhouette Score
-------------------------
2     | 0.0542
3     | 0.1084
4     | 0.1309
5     | 0.0959
6     | 0.0829
-------------------------
Highest k for Car: 4 (Score: 0.1309)
Observation: Scores are extremely low (< 0.14).
Reason: The data is uniformly distributed (factorial design), so no distinct clusters exist.
```

We repeated the same protocol:

- k=2: Silhouette ≈ 0.072

- k=3: Silhouette ≈ 0.108
- **k=4: Silhouette ≈ 0.131 (Highest, but negliable)**
- k=5: Silhouette ≈ 0.096
- k=6: Silhouette ≈ 0.088

Discussion:
The results for the Car dataset are uniformly poor, with all Silhouette scores below **0.14**.

1. **Lack of Structure:** The Car dataset comes from a "full factorial design," meaning almost every combination of features exists in the dataset. There are no natural "clumps" or "groups" of data points; the data is spread uniformly across the space.
2. **Dimensionality Issues:** After One-Hot Encoding, the data becomes high-dimensional and sparse. In such a binary space, the Euclidean distance between points tends to equalize (the "Curse of Dimensionality"), making K-means ineffective.
3. **Conclusion:** K-means is not a suitable algorithm for this specific categorical dataset because the data does not form spherical clusters.

1.4 General Conclusion

Comparing the two experiments:

1. **Numerical Data (Wine):** K-means successfully identified the correct number of clusters (

k = 3), though the separation was moderate.

2. **Categorical Data (Car):** K-means failed to find meaningful patterns. The uniform distribution of the categorical combinations resulted in near-zero Silhouette scores.